

UNIVERSITY OF ASIA PACIFIC
DEPARTMENT OF CSE

Lab Report - 1

COURSE CODE: CSE 402.

COURSE TITLE: OPERATING SYSTEM LAB.

SUBMITTED BY:

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SEMESTER : 4th YEAR 1st SEMESTER

SECTION : C - 2

DATE : 02 - 08 - 2026

SUBMITTED TO

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1. Problem:

Implement the **First Come First Serve (FCFS)** CPU Scheduling Algorithm for the following processes.

Process	Arrival Time (AT)	Burst Time (BT)
P0	3	1
P1	5	3
P2	2	2
P3	1	2
P4	6	3

Calculate CT, TAT, WT, AVG TAT and AVG WT. Also represent the process execution sequence.

2. Source Code:

FCFS.py

```python
lst = [
    ["P0",3,1],
    ["p1",5,3],
    ["p2",2,2],
    ["p3",1,2],
    ["p4",6,3]
]

lst.sort(key=lambda x:x[1])

ct = []
tat = []
wt = []
current_time = 0

for i in lst:
    name, at, bt = i

    if current_time < at :
        current_time = at

    current_time += bt
    ct.append(current_time)

    turnaround = current_time - at
    tat.append(turnaround)

    waiting = turnaround - bt
    wt.append(waiting)
```

```
32
33  for i in range(len(lst)):
34      print(lst[i][0], "\t",
35            lst[i][1], "\t",
36            lst[i][2], "\t",
37            ct[i], "\t",
38            tat[i], "\t",
39            wt[i], "\t")
40
41  avg_tat = sum(tat) / len(tat)
42  print("Average Turnaround time: ",avg_tat)
43
44  avg_wt = sum(wt)/ len(wt)
45  print("Average Waiting time: ", avg_wt)
46
47
```

3. Output:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER

PS C:\Users\hp\Desktop\CSE 402 Lab\Lab 2> & "C:\Program Files\Pyth
P_id    AT      BT      CT      TAT     WT
p3      1       2       3       2       0
p2      2       2       5       3       1
P0      3       1       6       3       2
p1      5       3       9       4       1
p4      6       3       12      6       3
Average Turnaround time:  3.6
Average Waiting time:  1.4
PS C:\Users\hp\Desktop\CSE 402 Lab\Lab 2>
```

4. Discussion:

In this experiment, the FCFS CPU Scheduling Algorithm was implemented using Python. The processes were first sorted according to their arrival times. Each process was executed in the order of arrival without preemption. After execution, the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) were calculated for every process. The average turnaround time and average waiting time were also computed. The results show that FCFS is easy to implement and works efficiently when processes arrive in a simple sequence. However, long processes can increase the waiting time of the processes arriving later, which is known as the convoy effect.

5. Conclusion:

The FCFS scheduling algorithm was successfully implemented and tested. The execution order followed the arrival times of the processes, and all scheduling parameters were calculated correctly. The experiment helped in understanding how CPU scheduling works and how FCFS determines Completion Time, Turnaround Time, and Waiting Time. Although FCFS is simple and fair in terms of arrival order, it may not provide the best average waiting time when longer processes execute before shorter ones.